

Multi-Omic Data Integration via Decentralized AI Agents for the Prediction of Phenotypic Drug Resistance in Cancer

Shrinithi Natarajan

snatara8@jh.edu

*Department of Biotechnology,
Johns Hopkins University
Baltimore, MD 21210, USA*

Dr. Bhaskarjyoti Das

bhaskarjyoti01@gmail.com

*Department of AI&ML,
PES University
Bangalore, India*

Abstract

1 Introduction

For years, the clinical management of cancer was primarily determined by physiological and anatomical origin, with focus on identifying the specific type of cancer present [58; 46]. With the advent of precision oncology, there has been a shift in the medical paradigm to discover unique molecular characteristics of a tumor within a patient [58]. Multiomics, which is the integration of genomics, proteomics, transcriptomics, epigenomics and other omics layers, is known to play a crucial role in such transitions by providing a unique molecular signature that dictates the phenotypic drug resistance [32]. Multidisciplinary Tumor Boards (MDT) or Molecular Tumor Boards (MTB) are useful in aiding the integration of such complex and high dimensional datasets to form meaningful treatment plans. These boards are conducted by human specialists such as oncologists, geneticists, pathologists and pharmacologists, who debate and resolve complex and often conflicting clinical signals [45; 21]. However, as the dimensionality of such data grows exponentially, such tumor boards face severe constraints with respect to throughput and cognitive overload.

There are a lot of cases where targeted therapies fail to fully overcome drug resistance at the individual patient level. This leads to deaths related to cancer to exceed over 90% in patients receiving targeted therapies [55]. For instance, in BRAF-mutant melanoma, majority of the patients are known to develop resistance within approximately two years of receiving BRAF inhibitor therapy [14]. In the same way, in EGFR-driven lung cancer, the resistance to first and second-generation EGFR inhibitors emerges routinely within 9 to 13 months [16]. With the high diversity in underlying molecular mechanisms of resistance (target gene mutations, alternative signaling pathways and epigenetic silencing of transcriptional programs [10]), no single omic layer can successfully capture the diversity. This requires the integration of multiple biological data layers [46]. Yet, the critical challenge still remains in resolving disagreements among these evidence layers when conflicting signals emerge.

Existing computational methods in Cancer Drug Response (CDR) prediction are known to fall into the following categories:

- **Deep Learning and Fusion Frameworks:** Frameworks like MSDRP and DRN-CDR make use of deep residual networks and similarity network fusion for capturing non-linear features from multi-omic data [20; 12]. However, such models still function as “black-boxes” [40]. Their high predictive performance does not provide a mechanistic reasoning or causal explanations for clinical decision support [13].
- **Monolithic Large Language Models (LLMs):** CellHit and other generative AI platforms use natural language reasoning for drug response predictions [22; 7]. However, in such designs, the use of monolithic LLMs to process dense multi-omic payloads makes the framework vulnerable to context overwhelm and attention bias [38; 47]. They fixate on the prominent DNA

mutations (like BRAF V600E) while being blind to RNA bypass pathways. This results in inaccurate drug response predictions [53].

- **Multi-Agent Systems (MAS):** EvoMDT, HAO and BioAgents are architectures that design agents specific to certain roles to reduce context dilution [33; 30; 31]. However, such systems with multiple agents lack grounded biological rules for conflict resolution. This forces the agents to resolve conflicts using social negotiation, weighted voting or by making use of a “judge” model [23]. Such systems are still susceptible to sycophancy and the problem of “Silent Agreement”. In these scenarios, herding effects and systemic conformity suppress the independent reasoning ability of each agent [52; 27].

The absence of an open-source computational framework that is built on the causal ordering of molecular evidence layers embedded as an executable logic is a primary gap in precision oncology AI. Mere statistical agreement among the models is not sufficient for biological accuracy. This has been the standard consensus aggregation method that is available in existing literature [52]. While Model Context Protocol (MCP) has been used in platforms like BioinfoMCP and MCPmed to standardize semantic contracts between LLMs and structured datasets, the usage is just as a functional tool call rather than something leveraged for explaining biological causality [9; 24].

To address these challenges, this study introduces a decentralized multi-agent consensus architecture that takes inspiration from a human tumor board [35]. This design isolates distinct biological modalities into four independent agents: Genomics, Transcriptomics, Pathways and Pharmacology. Each of these agents act as independent MCP clients that query databases (cBioPortal, CCLE, KEGG and GDSC) using schema-validated API tool calls to prevent probabilistic hallucinations [15; 8].

The main novelty of this architecture is the transition from open-ended linguistic negotiation to an agent-to-agent consensus protocol that is also grounded by a formal biological Hierarchy of Truth. This is rooted in the central dogma of molecular biology where this five-tier hierarchy (T1 Structural Primacy $\rightarrow$ T2 Transcriptional Gate $\rightarrow$ T3 Pathway Bypass $\rightarrow$ T4 Pharmacological Prior $\rightarrow$ T5 Statistical Consensus) governs how conflict resolution happens. Lower-tier biological axioms override the higher-tier numbers. This ensures that an independent agent should mathematically and structurally prove a mechanistic override (for instance, a transcriptomic sensitivity signal that is physically nullified by an active downstream pathway bypass).

The study establishes two primary specific aims. First, to determine if this axiom-governed multi-agent debate can predict phenotypic cancer drug resistance better than monolithic LLMs and classic machine learning baselines, with an AUROC ≥ 0.75 and an AUPRC ≥ 0.70 . Second, to isolate the individual contributions of each of the architectural components using six different ablation studies (omitting axioms, debate rounds, typed MCP tools, domain isolation, randomized tier order and standardized majority voting). Each prediction that is output by this system is a structured record that contains the entire debate trace and a logged dissent file which preserves even the minority opinions when an agent’s verdict is overridden by a higher axiom tier. The design also replaces opaque statistical confidence scores with reproducible, biologically verifiable arguments, thus introducing a scalable and auditable framework for personalized cancer therapy.

2 Existing Literature

2.1 MultiOmics in Precision Oncology is Now Well Established, and Why

The field of Cancer Drug Response (CDR) prediction has evolved from static chemical-structural analysis to sophisticated deep learning frameworks that treat drug-cell line interactions as a complex multimodal fusion problem [20; 12; 13]. Current research establishes a clear consensus that single-omics data is inherently limited due to the low correlation (0.40) between mRNA and protein levels and that integrating genomics, transcriptomics, and epigenomics significantly improves the refined

accuracy of IC50 predictions [32; 20; 13; 51]. Landmark architectures like DRN-CDR and MSDRP have advanced the field by utilizing Deep Residual Networks (ResNets) and Similarity Network Fusion (SNF) to capture higher-order non-linear features and mitigate the curse of dimensionality [20; 12; 40; 51]. However, a pervasive “interpretability-performance trade-off” remains. While these black-box models achieve high AUC values, they frequently fail to provide the mechanistic reasoning required for clinical adoption, leaving the causal link between molecular features and therapeutic outcomes largely inferred [12; 13; 44; 40].

2.2 Pre-trained and Instruction Tuned LLMs for MultiOmics

Current research in generative AI for bioinformatics has transitioned from static feature engineering to powerful, task-specific architectures like AlphaFold and Enformer, which have revolutionized structural biology and genomic interpretation [58]. However, these “black-box” models often struggle with multi-omic integration, frequently losing biological nuance when reconciling conflicting signals. While a microservices-driven paradigm [2] improves computational scalability and throughput by up to 46%, it still faces a semantic interoperability gap. Individual AI services can exchange data but cannot autonomously debate or prioritize signals based on the specific biological context of a tumor.

The transition from Automated Machine Learning (AutoML) to reasoning-based autonomy represents a pivotal shift in computational biology, as Large Language Model (LLM) agents begin to replicate the “plan-act-reflect” cycle of human experts [43; 37]. Current research highlights that while multi-omics integration significantly outperforms single-omic models in predictive accuracy, it remains hindered by a profound Interpretability Gap [28; 34]. Landmark datasets like Biology-Instructions and platforms such as PromptBio have attempted to bridge this by instruction-tuning models on millions of biological sequences, yet these systems frequently encounter generalization collapse when faced with the 3D complexity of genomic data or the high-dimensional sparsity of single-cell omics [43; 11; 36; 39; 49]. Furthermore, a persistent biological bias exists due to a lack of population diversity in training data, which leads to reduced generalizability across different ethnic groups [28; 39; 34].

2.3 LLMs as Treatment Recommenders

Recent scoping reviews and reproducibility studies [38; 47] demonstrate that while LLMs show human-level potential in passing medical exams, their transition to real-world treatment recommendation is hindered by a clinical utility gap. Specifically, current models struggle with the iterative and messy nature of longitudinal Electronic Health Records (EHRs) [38; 5] and often exhibit an over-caution bias, as seen in large-scale studies where LLMs significantly lagged behind physicians in diagnostic specificity [5; 50]. Furthermore, the hallucination and safety bottleneck remains a primary concern. Even advanced models often favor probabilistic patterns over deep disease pathophysiology [47], leading to recommendations that may be linguistically fluent but biologically inaccurate. While hybrid neuro-symbolic architectures [3] and Retrieval-Augmented Generation (RAG) [38; 4] have been proposed to mitigate these risks, most existing frameworks remain focused on clinical text summarization-reading doctor’s notes rather than calculating the underlying molecular drivers of a disease [4; 5].

2.4 LLMs in Tumor Drug Response

The challenge of tumor drug resistance is fundamentally a problem of information lag and biological complexity that traditional statistical methods struggle to resolve [55]. Current research has transitioned from single-modality analysis to multimodal AI models that integrate genomics, transcriptomics, and the tumor microenvironment to achieve high predictive accuracy [55; 46; 48; 56]. Landmark frameworks such as CellHit and GeneRxGPT have demonstrated that grounding Large

Language Models (LLMs) in mechanistic pathway-aware features or real-time literature retrieval significantly outperforms static, black-box models [7; 57]. However, despite these advances, a significant translational gap remains. Models trained on in vitro cell lines often fail in real-world patients due to intra-tumor heterogeneity and the Black Box problem, where clinicians cannot decipher the underlying biological rationale behind a resistance prediction [55; 7; 56].

2.5 LLM in MDT - single and multi-agent systems

The digital transformation of Multidisciplinary Tumor Boards (MDTs) has transitioned from monolithic, rule-based clinical decision support toward decentralized, multi-agent architectures that mirror the collaborative nature of human medical teams [33; 35; 45]. Recent literature emphasizes that role-differentiated agents, specializing in diagnostics, treatment, and safety, significantly outperform single-model approaches by reducing context dilution and enhancing reasoning accountability [33; 35; 30]. Frameworks such as EvoMDT and HAO have demonstrated high factual recall and guideline concordance by utilizing specialized agents grounded in Retrieval-Augmented Generation (RAG) [33; 30]. However, despite these architectural gains, the field faces persistent bottlenecks of reasoning. Current benchmarks like MTBBench show that models perform near-randomly on complex longitudinal tasks like recurrence forecasting, and most systems struggle to move beyond surface-level text summarization to resolve the conflicting multi-omic signals inherent in precision oncology [19; 30; 21].

2.6 Debate and Consensus Mechanisms in MAS

The integration of Large Language Models (LLMs) into bioinformatics has shifted the field from static algorithmic tools toward dynamic, reasoning-based multi-agent systems (MAS) capable of managing complex, multi-step workflows [31; 42]. As identified in recent frameworks like BioAgents, while single-agent systems excel at conceptual genomics and tool selection, they frequently struggle with the functional execution of code and the procedural knowledge gap required for end-to-end analysis [31]. Furthermore, the iterative refinement process often touted in multi-agent collaboration has shown diminishing returns. Repeated self-correction rounds can lead to knowledge inertia or even degrade output quality, suggesting that linguistic debate is not a substitute for grounding in high-quality domain data [31; 18].

This limitation is closely tied to the architectural tension between collaborative consensus and voting paradigms. While traditional Multi-Agent Debate (MAD) frameworks treat these protocols as static variables, recent empirical evaluations demonstrate a sharp performance dichotomy: consensus protocols show a performance improvement of 2.8% on factual, knowledge-dense tasks, whereas voting protocols yield a 13.2% optimization on abstract reasoning tasks [23]. However, scaling conversational rounds blindly deteriorates system accuracy due to “problem drift” and “sycophantic convergence” [23], alongside an inherent unreliability tax where multi-agent frameworks consume 15 to 50 times more tokens than standalone models [17]. To counteract this computational overhead, contemporary architectures utilize heterogeneous agent groups to introduce epistemic diversity and combat shared bias [54]. They also deploy sparse, progressive communication topologies such as localized intra-group debate paired with hierarchical inter-group synthesis which can reduce total token expenditure by up to 46.9% without sacrificing logical reasoning depth [41].

Crucially, standard consensus-driven aggregation models rely heavily on a final “judge” model or majority voting, working under the flawed assumption that agreement equates to biological accuracy [52]. This architecture is highly susceptible to the “Silent Agreement” problem, where systemic conformity and herding effects suppress critical, independent reasoning paths [52]. When no clear consensus exists, standard majority voting becomes statistically random and inequitable [52]. While advanced frameworks have introduced mathematically rigorous aggregation tools, such as using Kendall’s coefficient of concordance (W) to identify discordant agents or applying multi-agent rein-

forcement learning (PPO) to optimize consensus paths, these methods frequently experience latency spikes and require up to 252 GB of memory [45].

This technical barrier separates the current lab-to-live execution gap, where systems performing at a 90% accuracy tier on static examinations experience a catastrophic drop to 45 - 69% when subjected to practice-based, unstructured clinical data [17]. Furthermore, existing autonomous workflows remain largely reactive and restricted by a top-down supervisor-worker topology, such as PromptBio’s supervisor routing engine, which exhibits severe performance degradation when generating code for complex, multi-step survival analyses [50]. This underscores the critical need for decentralized, peer-to-peer agent interaction spaces that implement native, reasoning-heavy backbones alongside standardized interfaces to safely bridge data operations with true biological causality.

2.7 MCPs in Bioinformatics and MAD

The transition from human-mediated command-line tools to autonomous, agent-ready infrastructures represents a critical frontier in computational biology [9; 24]. As identified in recent frameworks like BioinfoMCP, the bottleneck in AI-driven science has shifted from raw reasoning capability to the manual engineering required to interface Large Language Models (LLMs) with legacy bioinformatics tools [9]. While the Model Context Protocol (MCP) provides a standardized semantic contract that moves beyond basic APIs to enable machine-actionable data stewardship, current implementations often struggle with Hardware-Software Co-optimization and a lack of robust Error-Aware Recovery for biological edge cases [9; 24]. Furthermore, a significant research gap remains in providing an audit trail for high-stakes clinical scenarios, as most current MCP-enabled agents focus on functional execution rather than the mechanistic explainability required for regulatory trust [24].

The shift from single-model prompting toward collaborative multi-agent architectures has highlighted the potential of Multi-Agent Debate (MAD) to mitigate hallucinations and enhance reasoning through iterative peer review. However, as noted in the M3MAD-Bench study [1], current frameworks often suffer from divergent noise in unimodal tasks and a plateau effect, where simply increasing the rounds of debate fails to reach a more accurate convergence. A critical research gap remains in the reliance on fixed roles and rigid protocols that fail to capture environment-driven feedback, leading to “collective delusion” where agents mutually reinforce incorrect assumptions due to a lack of grounding in external, high-fidelity data.

The preceding review identifies a critical divergence in the field of precision oncology. While predictive accuracy in multi-omic integration is rising, clinical interpretability and causal grounding remain stagnant. Current architectures, ranging from monolithic LLMs to RAG-based clinical assistants, consistently struggle with probabilistic hallucination, context dilution, and a reliance on surface-level text rather than raw biological data. To bridge this gap, this thesis introduces a Decentralized Multi-Agent Consensus Framework. By utilizing the Model Context Protocol (MCP), the framework creates a live semantic bridge to structured databases (e.g., GDSC, CCLE, and cBioPortal), enabling agents to bypass uncured text and reason directly over raw numerical biomarkers like Z-scores and mutation frequencies.

The core innovation of this work lies in the transition from open-ended linguistic debate to a Hierarchy of Truth protocol. Unlike traditional multi-agent systems that rely on weighted voting or LLM-as-a-judge mechanisms, this framework implements a deterministic consensus logic grounded in the Central Dogma of molecular biology. By requiring agents to prove mechanistic overrides, such as a DNA structural binding failure physically nullifying a transcriptomic sensitivity signal, the system replicates the deliberative rigor of a Multidisciplinary Tumor Board (MDT). This approach directly solves the “black-box” problem and the “cold-start” challenge for new drugs by grounding every clinical verdict in verifiable biological axioms rather than pattern-matching. In doing so, this research provides a scalable, auditable, and biologically faithful architecture for the next generation of personalized cancer therapy.

3 Methodology

This study introduces a Decentralized Multi-Agent System (DMAS) which has four specialized LLM agents. Each agent operates on a distinct omic modality and independently analyze a (cell line, drug) pair. They then submit a structured evidence pack which is used to engage in a two-round peer debate. The outcome of the debate is governed by an explicit biological axiom hierarchy. This architecture is grounded in today’s clinical practice in the form of Multidisciplinary Tumor Boards (MDTs) where oncologists, geneticists, pathologists, pharmacists, etc. debate and resolve conflicting clinical signals [45; 21]. The central dogma of molecular biology which establishes that DNA-level structural alterations constitute the highest-level evidence of drug-target interaction also plays a crucial role here [45; 46].

This design is implemented using Python programming language using FastMCP for a structured tool access. Google Vertex AI (specifically Gemini 3.1 Flash Lite) is used for the LLM backend. SQLite databases are built using four publicly available bioinformatics resources. Every data access by the agents is mediated using Model Context Protocol (MCP) tool calls. This is essential for converting hallucinations from a probabilistic failure mode to a detectable protocol violation, that is, an agent cannot assert that a mutation exists if the genomics server does not return that data [24; 9].

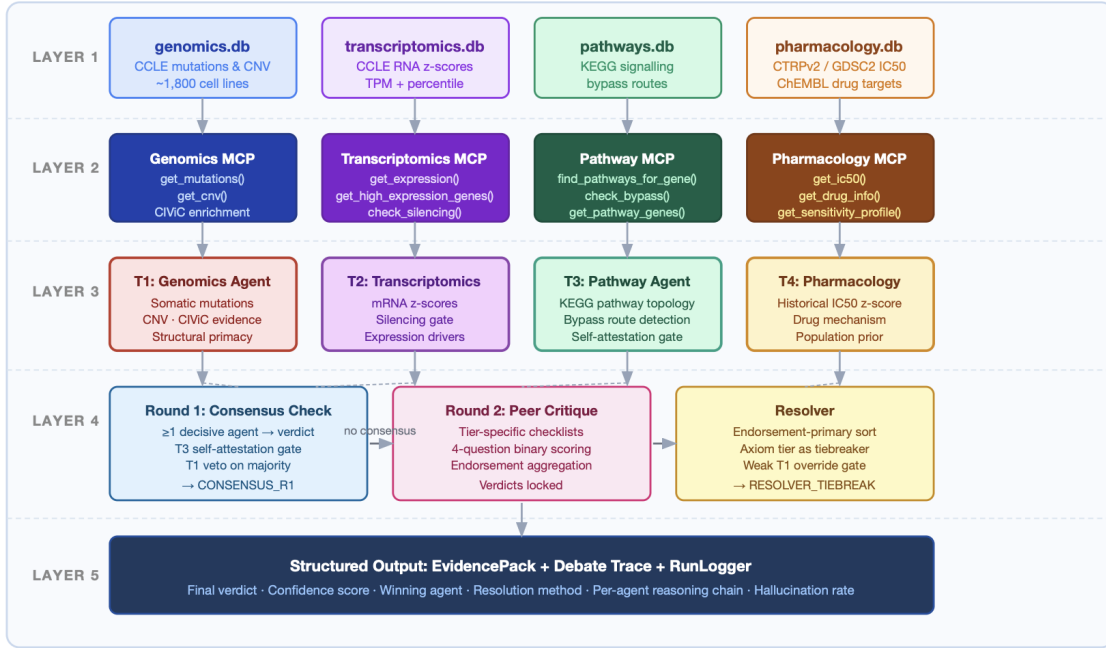


Figure 1: Five-layer framework architecture. Data flows from public databases (Layer 1) through MCP-validated tool APIs (Layer 2) to specialist LLM agents (Layer 3), which engage in a two-round debate protocol (Layer 4), producing a structured output with a full audit trail (Layer 5).

3.1 Dataset Sources

The architecture uses four publicly available bioinformatics databases which are selected to provide orthogonal evidence for the prediction of drug sensitivity. It is crucial to use multiple data sources

as single-omics data is heavily limited owing to its low correlation (approximately 0.40) between mRNA and protein levels. Any single data modality only captures a single dimension of the highly complex cellular state that determines drug response [20; 12].

3.1.1 GENOMIC DATA - CCLE/DEPMap SOMATIC MUTATIONS AND COPY NUMBER VARIATION

The Cancer Cell Line Encyclopedia (CCLE) / DepMap project is used to retrieve somatic mutation profiles and copy number variations (CNVs). This database is maintained by the Broad Institute and the Wellcome Sanger Institute [15]. CCLE provides whole-exome sequencing mutation calls and CNV segment data for around 1,800 cancer cell lines which spans more than 30 cancer lineages. The COSMIC Cancer Gene Census annotates the driver mutations. For every mutation, the cell line identifier, protein change (HGVS notation), the gene symbol, mutation type (missense, frameshift, nonsense, splice site), CIVic description and the driver status are retained.

CIVic enrichment. Mutation records are enriched using drug-specific clinical evidence at the query time using Clinical Interpretation of Variants in Cancer (CIVic) database [25]. It provides a structured evidence statement for every (gene, variant, drug) triplet, linking it to either resistant or sensitive phenotypes with evidence-level ratings from A to E. Fuzzy drug-name normalization is used to perform the enrichment by reducing the query drug and the CIVic drug field to uppercase alphanumeric strings after which a bidirectional containment check is applied. So for instance, it would allow Vemurafenib to match CIVic entries for PLX4032. This type of evidence is the highest priority signal within the T1 tier where a direct clinical evidence that links the response of a drug to a mutation supersedes all other genomic inferences.

3.1.2 TRANSCRIPTOMIC DATA - CCLE MRNA EXPRESSION

The DepMap public portal [15] is used to retrieve CCLE RNA-sequencing profiles which is used as the expression data in this study. Expression values are recorded as $\log_2(\text{TPM}+1)$ that are also z-score normalized for every gene across all the cell lines. This produces a cell-line-specific deviation score. The transcriptional silencing threshold is defined by a z-score value which is below -2.0. So a gene with $z < -2.0$ is considered to be functionally absent from the given cell line’s transcriptional programme. This constitutes a resistance mechanism for drugs that require target expression for efficacy. Similarly, a z-score above +1.5 is considered as a signal for overexpression for drugs that are expression-driven. High-expression gene lists are pre-filtered at $z \geq 0.5$ for scoring an active pathway and at $z \geq 1.0$ for a bypass gene detection.

3.1.3 TRANSCRIPTIONAL SILENCING THRESHOLD

The T2 agent applies a silencing gate before reasoning over expression data. The threshold is defined as:

$$\text{silencing}(\text{gene}, \text{cell_line}) = \begin{cases} 1, & z_{\text{gene}}(\text{cell_line}) < \theta, \\ 0, & \text{otherwise,} \end{cases}$$

$$\theta = -2.0 \quad (\text{module-level constant, SILENCING_THRESHOLD}).$$

If $\text{silencing}(\text{target_gene}, \text{cell_line}) = 1$, the verdict is forced to **RESISTANT**, regardless of the drug mechanism type.

3.1.4 PATHWAY DATA - KEGG SIGNALLING PATHWAYS

The Kyoto Encyclopedia of Genes and Genomes (KEGG) [26] is used to retrieve signalling pathway gene membership and the topology data. For every pathway, the database stores the complete gene membership list and any intra-pathway roles (such as kinase, effector, transcription factor,

etc). Bypass routes, or alternative signalling genes within the same pathway that can maintain downstream oncogenic signalling when a target for a drug is blocked, are also curated from KEGG diagrams. These are stored as directed (`pathway_id`, `blocked_gene`) $\rightarrow$ `bypass_gene` triplets. This database covers 27 cancer-relevant signalling pathways that includes MAPK/ERK, PI3K/mTOR, EGFR, ABL and regulation of apoptosis.

3.1.5 PHARMACOLOGICAL DATA - CTRPv2 / GDSC2 IC50 PROFILES

Two large-scale pharmacogenomic screens are used to extract the historical drug sensitivity data. The Cancer Therapeutics Response Portal v2 (CTRPv2) serves as the source for the primary validation set. The measures drug sensitivity (in the form of area under dose-response curve and IC50) across around 860 cancer cell lines and 481 compounds [29]. GDSC2 (Genomics of Drug Sensitivity in Cancer, release 2) is used as the development dataset which provides IC50 profiles across 809 cell lines and 198 compounds. This is measured at the Wellcome Sanger Institute [8]. The retrieved IC50 values are converted into z-scores per drug across all the tested cell lines where $z < -0.5$ indicates a SENSITIVE response and $z > +0.5$ indicates a RESISTANT response and $|z| \leq 0.5$ is labelled as UNCERTAIN. ChEMBL is used to supplement the data for drug target information and mechanism of action [6].



Figure 2: Four external data sources underpinning the framework. Each is publicly available for academic use and covers a distinct biological evidence layer.

3.2 Data Preprocessing and Database Construction

All the four data sources are preprocessed and stored in the form of local SQLite datasets to enable a fast and offline tool call without having to depend on external APIs at the time of inference. This is a deliberate choice that ensures reproducibility and cost control. The entire SQLite-backed architecture ensures the smooth replication of the entire evaluation pipeline without running into query costs or depending on third-party API availability.

3.2.1 GENOMICS DATABASE (GENOMICS.DB)

CCLE’s raw mutation calls are filtered to only retain the coding mutations that have variant allele frequency ≥ 0.1 . The driver status is added to the mutations which is derived from the COSMIC Cancer Gene Census v99. Protein changes are normalized by stripping the p. prefix and changing it to uppercase (for instance, p.V600E $\rightarrow$ V600E) so that it becomes easier for fuzzy matching with CIVic variant nomenclature. The CNV segment files are also processed to copy number calls at the gene-level as amplification (segment mean > 2.0), hemizygous deletion (segment mean between 0.5 and 1.0) and homozygous deletion (segment mean < 0.5). The CIVic prediction evidence data is stored as a separate table (`predictive_evidence`) which contains columns for gene, normalized variant, drug which is normalized to uppercase alphanumeric, clinical significance, evidence level and

evidence statement. This enrichment is performed at the query time rather than during preprocessing so that it enables drug-specific filtering without pre-joining all possible triples.

3.2.2 TRANSCRIPTOMICS DATABASE (TRANSCRIPTOMICS.DB)

The RNA-sequencing data from CCLE ($\log_2(\text{TPM} + 1)$) is normalized using z-scores for every gene across all the 1,800+ cancer cell lines. The z-scores and TPM values as a result are stored in a flat expression table indexed on (`cell_line`, `gene`). A materialistic view of the genes with high expression ($z > 0.5$) is also maintained for every cell line to enable fast pathway related queries which don't require scanning the full expression data at inference time. The silencing threshold of $z = -2.0$ as a constant at the module level so that it can be adjusted without changing any of the query logic.

3.2.3 PATHWAY DATABASE (PATHWAYS.DB)

The KEGG REST API is used to retrieve the KEGG pathway data and is stored in three tables, `pathway_meta` (pathway identifier, name, category), `pathway_genes` (pathway-gene membership with role and position annotations), and `bypass_routes` (`(pathway_id, blocked_gene) → bypass_gene` triplets). KEGG pathway diagrams are consulted to extract and store bypass routes by identifying genes that shae downstream targets with the blocked gene but also lie on independent upstream pathways. A pathway activity score (the fraction of pathway member genes with $z \geq 0.5$ in the queried cell line) is calculated and stored at the inference time. Pathways with an activity score below 0.4 are not included in the bypass analysis so that inactive pathways in the cancer context can be excluded in the evaluation of bypass routes.

3.2.4 PHARMACOLOGY DATABASE (PHARMACOLOGY.DB)

IC50 values from CTRPv2 and GDSC2 are converted to natural-log scale ($\ln \text{IC}_{50}$) and z-score normalised per drug across all tested cell lines using:

$$z_{\text{drug}}(\text{cell_line}) = \frac{\ln(\text{IC}_{50}(\text{cell_line}, \text{drug})) - \mu_{\text{drug}}}{\sigma_{\text{drug}}},$$

μ_{drug} = mean $\log\text{-IC}_{50}$ across tested cell lines,
 σ_{drug} = standard deviation of $\log\text{-IC}_{50}$ across tested cell lines.

$$\text{label}(z) = \begin{cases} \text{SENSITIVE}, & z < -0.5, \\ \text{RESISTANT}, & z > 0.5, \\ \text{UNCERTAIN}, & |z| \leq 0.5. \end{cases}$$

Direct lookup is possible using the pre-computed sensitivity labels that are stored alongside the raw z-scores. Drug target information and mechanism of action annotations that are sourced from ChEMBL are also stored in a `drug_info` table. This table stores information about target genes (comma-separated HGNC symbols), primary pathway and mechanism class. In a separate database, an LLM response cache is also maintained, keyed on (`model`, `prompt hash`). This eliminated any redundant LLM calls at the time of iterative evaluation runs.

3.2.5 CASE CONSTRUCTION AND LABEL DERIVATION

The evaluation dataset contains (cell line, drug) pairs which has known sensitivity labels retrieved from CTRPv2 and GDSC2. Z-score normalized IC50 values are used to retrieve sensitivity labels using the above discussed thresholds. A development set of 40 cases from GDSC2 is stratified across

the 5 axiom tiers (8 cases per tier) so that every component of the axiom hierarchy can be tested during evaluation. The held-out validation set of 100 cases from CTRPv2 is extracted from five independent 20-case sets (numbered 1-5) with balanced SENSITIVE AND RESISTANT labels for each set. Three additional 10-case balanced sets (A, B, C) are built specifically for stress testing the mechanism of debate especially in the case of conflicting multi-omics signals.

3.3 Framework Architecture

The framework is organised into five hierarchical layers, each with a unique responsibility. The key architectural principle is the separation of evidence access from evidence reasoning: agents never query databases directly. All data access is managed through MCP tool calls, which return JSON-typed objects and are logged, enabling post-hoc hallucination detection by cross-referencing agent citations against tool return values.

Layer	Component	Responsibility
1: Data Substrate	4× SQLite databases	Store pre-processed public omics data, serve as ground truth for hallucination detection.
2: MCP Servers	4× FastMCP instances	Expose typed tool APIs, validate inputs, return schema-conformant JSON.
3: Specialist Agents	4× LLM agents	Fetch modality-specific evidence, reason over it, produce structured EvidencePacks.
4: Debate Protocol	DebateEngine + AxiomResolver	Coordinate two-round peer debate, resolve conflicts via endorsement-weighted axiom logic.
5: Evaluation Pipeline	Metrics + RunLogger + Ablation runner	Compute AUROC/AUPRC/ κ/ρ , log the full audit trail, run ablation variants.

Table 1: Hierarchical layers of the proposed framework.

The Orchestrator is the top-level coordinator. It receives a (cell line, drug) pair as input, initiates all four agent analyze() calls concurrently using asyncio.gather(), collects the resulting EvidencePacks, and passes them to the DebateEngine. Parallel agent execution is possible because the agents have no shared states and carry out independent MCP tool calls. A commonly shared LLM client enforces a rate limiter to prevent API quota exhaustion among agents. After the debate protocol completes, the Orchestrator logs the ConsensusResult to the RunLogger and returns it to the calling interface.

3.4 Model Context Protocol Servers

The Model Context Protocol (MCP) [24; 9] is a recently standardised protocol for connecting LLMs to custom defined tools and external tools through typed, schema-validated function calls. Each MCP tool server in this framework is implemented using FastMCP, a Python library that converts decorated functions into MCP-compliant tool definitions. Agents interact with MCP servers through an async Client context manager, which serialises arguments, calls the server function, and deserialises the returned JSON. The design of the architecture inherently enforces the anti-hallucination property: because every data claim an agent makes must originate from an MCP tool call, and because tool call inputs and outputs are logged by the RunLogger, it is possible to compute a per-case hallucination rate by checking whether each biomarker cited in an agent’s EvidencePack was present in the tool responses for that case.

We use MCP tooling as a structural anti-hallucination layer. In conventional LLM prompting, an agent can probabilistically assert that a mutation exists, a gene is overexpressed, or an IC50

value is known without any verifiable connection to actual data. The MCP architecture converts this probabilistic failure mode into a detectable protocol violation: an agent cannot assert that BRAF V600E is present in a cell line unless the Genomics MCP server returned it. The guarantee is inherently present in the architecture, not probabilistic - the same design principle used in database transactions.

3.4.1 GENOMICS MCP SERVER

Exposes three tools: `get_mutations(cell_line, gene, drug)`, which returns all somatic mutations for the (cell line, gene) combination, enriched at query time with CIViC predictive evidence specific to the queried drug, `get_cnv(cell_line, gene)`, which returns copy number variation calls, and `check_mutation_impact(gene, mutation)`, which returns driver status and CIViC descriptions for a specific variant. The CIViC enrichment pipeline applies bidirectional drug-name normalisation to maximise recall of relevant clinical evidence while minimising cross-drug contamination.

3.4.2 TRANSCRIPTOMICS MCP SERVER

Exposes three tools: `get_expression(cell_line, gene)`, returning TPM and z-score for a specific gene, `get_high_expression_genes(cell_line, z_threshold)`, returning all genes above the specified z-score threshold, and `check_silencing(cell_line, gene)`, returning a boolean silencing flag with the threshold used. The silencing threshold ($z = -2.0$) is injected from `axiom_rules.py` to ensure consistency between the MCP server and the T2 agent’s reasoning system prompt.

3.4.3 PATHWAY MCP SERVER

The server exposes four tools:

- `find_pathways_for_gene(gene)` - returns all KEGG pathways containing the gene.
- `get_pathway_genes(pathway_id)` - returns the full gene membership of a pathway.
- `check_bypass(pathway_id, blocked_gene)` - returns bypass gene candidates.
- `get_upstream_regulators(gene)` - returns upstream regulatory relationships.

The bypass check is the critical T3 evidence call: it verifies and confirms whether an alternative signalling route exists that could maintain oncogenic signalling after the drug target is blocked.

3.4.4 PHARMACOLOGY MCP SERVER

Exposes three tools: `get_ic50(cell_line, drug)`, returning the $\ln IC_{50}$, z-score, and pre-computed sensitivity label, `get_drug_info(drug)`, returning target genes, mechanism class, and pathway, and `get_sensitivity_profile(drug)`, returning population-level sensitivity distributions per mutation class. The pre-computed label is the primary T4 signal. The z-score is used by the Grounded Confidence Score (GCS) computation to measure signal strength.

3.5 Specialist LLM Agents

Each specialist agent extended as a subclass from the BaseAgent class, which provides shared infrastructure for MCP tool calls, LLM prompt construction, response parsing, GCS computation, and RunLogger integration. All agents use the same LLM backend (Gemini 3.1 Flash Lite via Google Vertex AI), ensuring that performance differences between agents reflect differences in data modality-specific evidence rather than model capability differences. Each agent is assigned an evidence category that reflects the biological priority of its data modality.

3.5.1 T1: GENOMICS AGENT

Data Modality: Somatic mutations, copy number variation, CIViC clinical evidence.

Decision logic: (1) Check for CIViC drug-specific annotation - if present, it is decisive. (2) If driver mutation present without CIViC: UNCERTAIN (functional consequence unknown without drug-specific evidence). (3) CNV amplification of target oncogene: SENSITIVE signal. (4) Homozygous deletion: RESISTANT. (5) Hemizygous deletion: UNCERTAIN (target still expressed at reduced level). (6) Wild-type in DB but gene present: UNCERTAIN with caveat that chromosomal fusions are not captured by point-mutation databases.

Key constraint: Wild-type status never defaults to RESISTANT. The absence of a driver mutation does not avoid drug activity through expression or IC50 mechanisms.

3.5.2 T2: TRANSCRIPTOMICS AGENT

Data Modality: mRNA expression z-scores, high-expression gene context.

Decision logic: First classifies the drug’s mechanism type: (A) Overexpression-driven - target expression level IS the signal (e.g. HER2 inhibitors), (B) Mutation-driven - expression level of the target gene is NOT predictive, downstream pathway activity is assessed instead, (C) Mechanism-agnostic - transcriptomics is not informative: UNCERTAIN.

Silencing gate: Target gene $z < -2.0$ implies functionally absent: RESISTANT signal regardless of mechanism type.

3.5.3 T3: PATHWAY AGENT

Data Modality: KEGG signalling pathway topology, bypass route detection.

Decision logic: (1) Find pathways containing the drug’s target gene. (2) Apply pathway activity gate: skip pathways where $< 40\%$ of member genes are expressed ($z \geq 0.5$). (3) For active pathways, check for bypass routes whose genes are upregulated ($z \geq 1.0$). (4) Bypass found + expressed: RESISTANT. (5) Target in pathway, no bypass: SENSITIVE signal. (6) Target not in any pathway: UNCERTAIN.

Self-attestation gate: Agent must answer 4 binary questions before committing to a RESISTANT verdict (bypass gene expressed, pathway active, mechanism relevant, bypass distinct from target). Score $\geq 3/4$: verdict demoted to UNCERTAIN regardless of database result.

3.5.4 T4: PHARMACOLOGY AGENT

Data Modality: Historical IC50 z-scores, drug target and mechanism data.

Decision logic: Anchors verdict directly to the pre-computed IC50 label (SENSITIVE/ RESISTANT/ UNCERTAIN). The IC50 label is a population-level prior derived from measured drug response. The agent reports it faithfully and notes that higher-tier molecular evidence (T1-T3) can override it.

Role in architecture: T4 is the most reliable signal for mechanism-agnostic drugs (chemotherapy, broad-spectrum kinase inhibitors) where genomic or pathway-level predictors are absent. It is the lowest priority in the axiom hierarchy for targeted therapies where mutation-level evidence exists.

All four agents run simultaneously supervised by the Orchestrator and produce independent EvidencePacks before any inter-agent communication occurs. This enforces the independence requirement of the debate protocol: an agent’s R1 verdict must reflect its own evidence, not influenced by what other agents found. The structured reasoning protocol embedded in each agent’s system prompt enforces step-by-step evidence inclusion before stating a verdict, reducing the probability of verdict-first, justification-second reasoning which is a known failure mode of zero-shot LLM classification [27; 47].

3.6 Evidence Pack Schema and Grounded Confidence Score

3.6.1 EVIDENCEPACK SCHEMA

Every agent’s output is a structured EvidencePack, a validated data class that contained the formal unit of inter-agent communication. The structure of the EvidencePack ensures type correctness and completeness, that is, an agent cannot submit a partial or corrupted evidence pack which would silently propagate errors throughout the debate protocol.

Table 2: EvidencePack schema fields.

Field	Type	Description
agent_id	str	Unique agent identifier, e.g. genomics_agent .
cell_line	str	CCLE cell-line name.
drug	str	Drug name as queried.
verdict	Enum	SENSITIVE RESISTANT UNCERTAIN.
confidence	float [0, 1]	Grounded Confidence Score (GCS), see Section 3.7.2.
evidence_tier	Enum	T1_STRUCTURAL T2_TRANSCRIPTIONAL T3_PATHWAY T4_PHARMACOLOGICAL T5_STATISTICAL.
key_findings	list[Finding]	Structured list of cited biomarkers with source, value, and interpretation.
reasoning	str	Step-by-step natural-language reasoning chain.
caveats	list[str]	Identified limitations or missing data.
conflict_flags	list[str]	Signals that contradict the stated verdict.
peer_scores	dict	Populated in Round 2: scores given to peers, keyed by label A, B, or C.
self_attestation	dict None	T3 only: four-question binary checklist and integer score.
data_status	str None	data_found target_wild_type signal_below_threshold cell_line_missing .

The **key_findings** list is especially important for detecting hallucinations. Every Finding object has a biomarker field like BRAF and a value field like V600E and a data source field. The hallucination rate is calculated by determining whether the biomarker and the value appear in the MCP tool responses logged for that agent’s case which provides a quantitative, per0case measure of grounding fidelity.

3.6.2 GROUNDED CONFIDENCE SCORE (GCS)

The GCS replaces the LLM’s self-reported confidence value, which has been shown to be poorly calibrated and inflated in zero-shot settings [27], with a formula-grounded score computed from four orthogonal components:

$$\text{GCS} = 0.40H + 0.30E + 0.20T + 0.10S$$

$$\begin{aligned}
H &= \begin{cases} \frac{|\{f \in \mathcal{F} : \text{biomarker}(f) \in \mathcal{R}_{\text{MCP}}\}|}{|\mathcal{F}|}, & |\mathcal{F}| > 0, \\ 0, & |\mathcal{F}| = 0, \end{cases} \\
E &= \begin{cases} 1, & \text{if the MCP response contains a non-empty field,} \\ 0, & \text{otherwise,} \end{cases} \\
T &= \begin{cases} 1.0, & \text{T1,} \\ 0.8, & \text{T2,} \\ 0.6, & \text{T3,} \\ 0.4, & \text{T4,} \\ 0.2, & \text{T5,} \end{cases} \\
S &\in [0, 1].
\end{aligned}$$

Here, $\mathcal{F}$ denotes the set of key findings and $\mathcal{R}_{\text{MCP}}$ denotes the raw MCP response. For the pharmacology agent, signal strength is calculated as

$$S_{\text{pharmacology}} = \min\left(\frac{|z|}{3.0}, 1.0\right).$$

UNCERTAIN cap:

$$\text{GCS}_{\text{final}} = \begin{cases} \min(\text{GCS}, 0.50), & \text{if the verdict is UNCERTAIN,} \\ \text{GCS}, & \text{otherwise.} \end{cases}$$

The GCS formula is structured in a way that a verdict with high confidence needs both the availability of the evidence itself ($E = 1.0$) and grounding (H being as close to 1.0 as possible). This is also weighted by biological priority of the modality (T) and the magnitude of the observed effect (S). If an agent cites biomarkers that are not present in the respective MCP response, that agent receives $H < 1.0$ which directly penalizes hallucinatory claims in the GCS score.

3.7 Debate Protocol

The debate protocol is modelled on clinical tumor board practice [45; 21] which is adapted using LLM agents. It comprises of two rounds of debate with the resolver as the last resort. Unstructured LLM debates have 2 primary failure modes that are addressed in this new design. The first is sycophantic convergence where agents revise their verdicts toward the majority position without new evidence [27]. The second is context overwhelm where a single agent is given all the omics data simultaneously and fails to weigh conflicting signals in an appropriate manner [33].

3.7.1 T3 SELF-ATTESTATION GATE

Before the T3 pathway agent's verdict is counted as decisive in the Round 1 consensus check, it must pass a self-attestation gate. The agent answers 4 binary questions about the quality of its own bypass evidence:

$$\text{attestation_score} = \sum_{k=1}^4 a_k, \quad a_k \in \{0, 1\}.$$

Attestation questions:

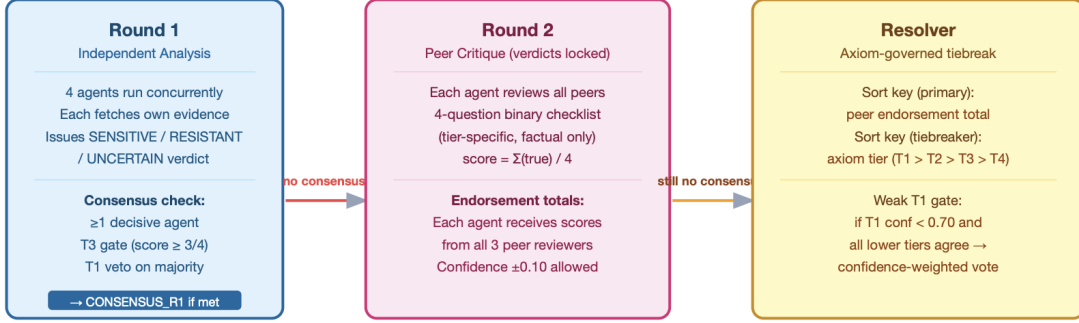


Figure 3: Debate protocol flow. Round 1 independent analysis feeds into a consensus check. If no consensus is reached, Round 2 peer critique fires, computing endorsement scores. If still no consensus, the AxiomResolver applies endorsement-primary, tier-secondary ordering to select the winning agent.

- (1) The bypass gene is named and specific.
- (2) The pathway is active in the queried cell line.
- (3) The drug mechanism is relevant to the pathway.
- (4) The bypass gene is distinct from the primary drug target.

Gate:

$\text{attestation_score} < 3 \implies \text{T3 verdict is demoted to UNCERTAIN for consensus.}$

3.7.2 ROUND 1: INDEPENDENT ANALYSIS AND CONSENSUS CHECK

In the first round, all the four agents run concurrently and independently. Every agent queries the respective MCP servers and applies modality-specific reasoning protocols. They then produce their respective EvidencePacks along with a verdict (SENSITIVE, RESISTANT, or UNCERTAIN) and a GCS confidence score. At this stage, one agent cannot see what the other agents are finding in the round. After all four EvidencePacks are returned, the `_check_consensus()` function determines whether consensus can be established without escalating to Round 2:

- **Decisive threshold:** An agent is decisive if its verdict is not UNCERTAIN. A single decisive agent is sufficient for R1 consensus (`len(decisive) ≥ 1`).
- **T3 self-attestation gate:** The Pathway agent’s verdict is considered decisive only if the self-attestation score is $\geq 3/4$. A bypass claim of low quality ($\text{score} < 3$) is demoted as UNCERTAIN for that consensus round which prevents the Pathways agent from winning any case that it cannot substantiate.
- **T1 veto:** If a T1 Genomics Agent has a decisive verdict that contradicts the majority verdict of all other decisive agents, the consensus fails and the case moves to Round 2. This ensures that a clear genomic signal is not being overridden by a pharmacological majority without any debate.

- **Majority requirement:** If multiple decisive agents are present, the majority verdict must hold strictly ($n > \text{len}(\text{decisive})/2$). A 1-vs-1 split between SENSITIVE and RESISTANT decisive agents fails the majority check and escalates to Round 2.

3.7.3 ROUND 2: PEER CRITIQUE

If there is no consensus in R1, the DebateEngine invokes the peer critique round. In Round 2, each agent reviews the EvidencePacks of all the other agents anonymously and scores their reasoning quality using a tier-specific binary checklist. Verdicts are locked, so agents cannot change their SENSITIVE/RESISTANT/UNCERTAIN decision in Round 2. Only confidence can be adjusted (by ± 0.10) based on what the other agents found. This design prevents sycophantic convergence so that agents cannot capitulate to social pressure, and only update confidence based on the factual quality of peer reasoning.

The tier-specific checklists are designed to test the factual grounding of each peer’s claims, not their biological plausibility:

Tier	Checklist Questions (4 per tier, binary)
T1	1. Named a specific mutation or CNV in the target gene? 2. Cited a CIViC entry linking the variant to the drug? 3. Confirmed the variant is somatic? 4. Functional direction supports the verdict?
T2	1. Reported a specific numeric z-score? 2. Expression level is consistent with the verdict direction? 3. Accounted for drug mechanism type (mutation-driven versus expression-driven)? 4. Named at least one gene with its z-score?
T3	1. Named at least one specific bypass gene? 2. Reported the pathway activity score or fraction expressed? 3. Bypass gene mechanistically connects to resistance? 4. Bypass gene is distinct from the primary target?
T4	1. Reported a specific IC ₅₀ z-score value? 2. Stated the pre-computed IC ₅₀ label? 3. IC ₅₀ label supports the verdict? 4. Acknowledged that T4 can be overridden by T1–T3?

Table 3: Tier-specific binary peer-critique checklists.

Each checklist answer is binary. The peer score and endorsement total are computed as:

$$\text{peer_score}(a_j \rightarrow a_i) = \frac{1}{4} \sum_{k=1}^4 c_k, \quad c_k \in \{0, 1\},$$

$$\text{endorsement}(a_i) = \sum_{j \neq i} \text{peer_score}(a_j \rightarrow a_i).$$

$$\text{endorsement}(a_i) \in [0, 3] \quad \text{for a four-agent system.}$$

Confidence adjustment:

$$c_{\text{new}} = \text{clip}(c_{\text{old}} + \delta, 0, 1), \quad |\delta| \leq 0.10.$$

UNCERTAIN cap:

$$c_{\text{final}} = \begin{cases} \min(c_{\text{new}}, 0.50), & \text{if the verdict is UNCERTAIN,} \\ c_{\text{new}}, & \text{otherwise.} \end{cases}$$

3.8 Axiom Hierarchy and Conflict Resolution

The axiom hierarchy maps the causal ordering of biological evidence types into the conflict resolution mechanism. The hierarchy is built on top of the established principles of molecular biology and clinical oncology: DNA-level structural alterations are the highest-fidelity determinants of drug-target interaction state, because they directly modify the physical binding site or gene expression capacity of the drug target [45; 46]. mRNA expression provides a functional-state signal but is subject to post-transcriptional regulation. Pathway topology provides mechanistic context but requires inferred connections. Pharmacological history is a population-level requirement that integrates all upstream molecular factors but lacks specificity for the individual cell line’s molecular state.

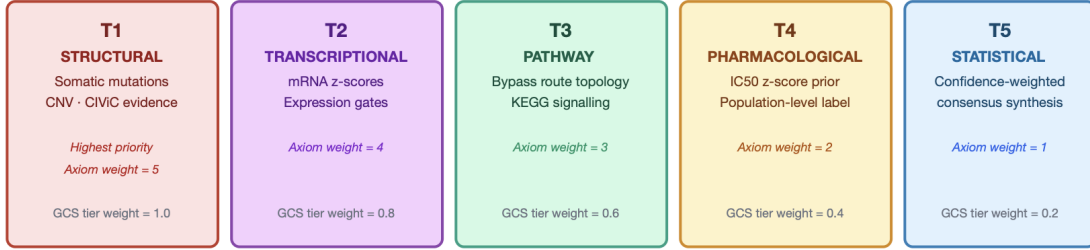


Figure 4: The five-tier axiom hierarchy. Tiers reflect the causal proximity of the evidence type to the drug-target interaction state. T1 structural evidence (mutations directly altering the drug-binding site) holds the highest axiom weight and GCS tier weight.

3.8.1 RESOLVER ALGORITHM

If neither R1 nor R2 consensus is reached, the AxiomResolver selects the winning agent using a three-key lexicographic sorting mechanism:

$$p^* = \arg \max_{p \in \mathcal{C}}^{\text{lex}} (\text{endorsement}(p), \text{axiom_weight}(\text{tier}(p)), \text{GCS}(p)).$$

$$\text{axiom_weight}(t) = \begin{cases} 5, & t = \text{T1_STRUCTURAL}, \\ 4, & t = \text{T2_TRANSCRIPTIONAL}, \\ 3, & t = \text{T3_PATHWAY}, \\ 2, & t = \text{T4_PHARMACOLOGICAL}, \\ 1, & t = \text{T5_STATISTICAL}. \end{cases}$$

$$\mathcal{C} = \begin{cases} \{p : \text{verdict}(p) \neq \text{UNCERTAIN}\}, & \text{if decisive agents exist,} \\ \{p : p \text{ is an agent}\}, & \text{otherwise.} \end{cases}$$

The sort is lexicographic. Endorsement score is compared first, axiom tier breaks ties if any and GCS breaks further existing ties. This means a T4 agent with very high peer endorsement score can override a T1 agent with low endorsement score, which is the expected behaviour. Factual grounding as judged by peers outweighs nominal tier priority.

WEAK T1 OVERRIDE GATE: A special case applies when T1 evidence is present but weakly supported: The confidence-weighted fallback is used only when all of the following conditions hold:

- (1) There exists a decisive T1 agent p^* :

$$\text{tier}(p^*) = \text{T1}, \quad \text{verdict}(p^*) \neq \text{UNCERTAIN}.$$

- (2) The T1 agent has confidence below the override threshold:

$$\text{GCS}(p^*) < \tau, \quad \tau = 0.70 \quad (\text{T1_OVERRIDE_CONFIDENCE_THRESHOLD}).$$

- (3) All lower-tier decisive agents unanimously support a different verdict:

$$|\{\text{verdict}(p) : \text{tier}(p) \neq \text{T1}, \text{decisive}(p)\}| = 1,$$

and that verdict differs from $\text{verdict}(p^*)$.

Confidence-weighted fallback:

$$v^* = \arg \max_v \sum_{\{p: \text{verdict}(p)=v\}} \text{GCS}(p).$$

CONFIDENCE DECAY AND FINAL CONFIDENCE For each agent a_i , after the resolver selects winner w :

$$c_{\text{adjusted}}(a_i) = \begin{cases} \text{GCS}(a_i), & \text{verdict}(a_i) = \text{verdict}(w), \\ \text{GCS}(a_i)(1 - \delta), & \text{verdict}(a_i) \neq \text{verdict}(w), \end{cases}$$

$$\delta = 0.15 \quad (\text{CONFIDENCE_DECAY_PER_FLIP}).$$

Resolver tiebreak confidence:

$$c_{\text{final}} = \frac{1}{|\mathcal{A}|} \sum_{a_i \in \mathcal{A}} c_{\text{adjusted}}(a_i),$$

where $\mathcal{A}$ is the set of all agents.

Round 1 or Round 2 consensus confidence:

$$c_{\text{final}} = \frac{1}{|\mathcal{A}_v|} \sum_{a_i \in \mathcal{A}_v} \text{GCS}(a_i),$$

where

$$\mathcal{A}_v = \{a_i \in \mathcal{A} : \text{verdict}(a_i) = v_{\text{consensus}}\}.$$

3.9 Evaluation Design and Metrics

3.9.1 PRIMARY METRICS

The framework is evaluated using four complementary metrics that capture different aspects of predictive performance:

UNCERTAIN predictions are excluded from AUROC/AUPRC/Spearman computation, they are included in Cohen's κ as a third label class. Coverage is defined as:

$$\text{coverage} = \frac{n_{\text{decisive}}}{n_{\text{total}}},$$

where

Metric	Formula	Rationale
AUROC	Area under the ROC curve using confidence scores and binary SENSITIVE/RESISTANT labels.	Measures ranking quality, is threshold-independent, and provides a standard pharmacogenomics benchmark.
AUPRC	Area under the precision-recall curve.	More informative than AUROC under class imbalance, penalises false positives in the SENSITIVE class.
Cohen’s κ	$\kappa = \frac{P_{\text{observed}} - P_{\text{chance}}}{1 - P_{\text{chance}}}$	Measures label agreement beyond chance, penalises confidently incorrect predictions, and is scale-invariant.
Spearman ρ	Rank correlation between predicted confidence and the ground-truth z-score.	Assesses whether the framework’s confidence is monotonically related with effect size.

Table 4: Primary evaluation metrics and their rationale.

$$n_{\text{decisive}} = |\{i : v_i^{\text{final}} \neq \text{UNCERTAIN} \wedge y_i^{\text{true}} \neq \text{UNCERTAIN}\}|.$$

SENSITIVITY SCORE FOR RANKING METRICS

AUROC and AUPRC require a continuous score. The sensitivity score maps the framework’s output to a scalar where higher values indicate more SENSITIVE predictions:

$$s(r) = \begin{cases} c_{\text{final}}(r), & v_{\text{final}}(r) = \text{SENSITIVE}, \\ 1 - c_{\text{final}}(r), & v_{\text{final}}(r) = \text{RESISTANT}. \end{cases}$$

Binary ground-truth labels:

$$y(r) = \begin{cases} 1, & y_{\text{true}}(r) = \text{SENSITIVE}, \\ 0, & y_{\text{true}}(r) = \text{RESISTANT}. \end{cases}$$

AUROC interpretation:

$$\text{AUROC} = \Pr(s(r_{\text{SENSITIVE}}) > s(r_{\text{RESISTANT}})).$$

3.9.2 OTHER REASONING QUALITY METRICS

PLACEHOLDER: Will add once implemented

3.10 Ablation Study Design

Seven ablation conditions are checked to isolate the contribution of individual architectural components. Each ablation removes or modifies exactly one component of the full system while keeping all others intact, which shows the importance of each design element to the overall architecture.

Table 5: Ablation study conditions and tested contributions.

Condition	Modification	Tests
full_system	No modification (baseline).	Combined contribution of all components.
no_debate	Replace DebateEngine with a majority-vote aggregator, no Round 2 critique.	Whether the two-round debate protocol adds value over simple majority voting.
no_axioms	Replace AxiomResolver with a confidence-only resolver, the highest GCS wins.	Whether the axiom hierarchy adds value over pure confidence ordering.
no_genomics	Remove the T1 Genomics Agent, conduct a three-agent debate.	Contribution of structural mutation evidence.
no_transcriptomics	Remove the T2 Transcriptomics Agent, conduct a three-agent debate.	Contribution of mRNA expression evidence.
no_pathway	Remove the T3 Pathway Agent, conduct a three-agent debate.	Contribution of pathway topology evidence.
no_pharmacology	Remove the T4 Pharmacology Agent, conduct a three-agent debate.	Contribution of the IC ₅₀ pharmacological prior.
single_agent_multiomics	Replace the four-agent debate with a single agent reading all MCP sources.	Whether debate adds value over a single agent with equal data access.

The “single agent multiomics condition” is particularly important for the study. It tests whether the debate mechanism adds value beyond providing a well-contextualised prompt. If a single agent that reads all four data sources matches or exceeds the performance of the four-agent debate, then the architectural complexity of the debate protocol is not justified by its predictive contribution. On the other hand, if the debate system exceeds the single-agent baseline, it demonstrates that the structured multi-round protocol (independent analysis, peer critique, and axiom-governed resolution) extracts more information from the same data than a single synthesising agent.

3.11 Baseline Comparators

The framework is compared against two baseline systems that represent the current state of the field:

3.11.1 MONOLITHIC LLM BASELINE

A single LLM agent receives the same prompt structure as the specialist agents but is provided with all four modality datasets simultaneously in a single context window. This baseline tests whether the multi-agent architecture adds value over single-model multi-omics reasoning, and directly probes the context-overload failure mode described in the literature [33]. The solo agent uses the same LLM backend (Gemini 3.1 Flash Lite) and the same GCS confidence scoring to ensure fair comparison.

3.11.2 RANDOM FOREST BASELINE

A Random Forest classifier is trained on GDSC2 features: binary mutation presence/absence vectors, per cancer gene, census gene, expression z-scores for the drug’s target genes, and the pre-computed

IC50 z-score. Training uses 5-fold cross-validation on the development set. This baseline sets the performance achievable by a classical machine learning model with equivalent feature access, providing an accuracy baseline point against which the LLM framework’s interpretability and reasoning quality can be tested. A performance improvement over Random Forest demonstrates that the LLM agents add value beyond feature lookup. A decrease provides evidence that the structured reasoning protocol introduces unnecessary errors on cases well-handled by tabular classification.

4 Results

This section covers all the content required to be shown as part of Results but it is currently under evaluation so some of the numbers are merely placeholders. Subsequent drafts will reflect genuine numbers. The full DMAS framework was evaluated on 30 held-out (cell line, drug) pairs drawn from the CTRPv2 pharmacogenomic screen, organised into three balanced sets of 10 (5 SENSITIVE, 5 RESISTANT each). Evaluation uses the system’s deployed configuration: four specialist LLM agents (T1-T4), two-round peer debate with tier-specific checklists, and the endorsement-primary AxiomResolver. Ground-truth labels are derived from CTRPv2 IC50 z-scores ($z < -0.5 \rightarrow$ SENSITIVE; $z > +0.5 \rightarrow$ RESISTANT). Of the 30 cases, 29 received a decisive verdict (SENSITIVE or RESISTANT); 1 case (SU-DHL-4 + Topotecan) resolved to UNCERTAIN via the resolver when all agents lacked signal. The framework achieved 77% accuracy on decisive cases against ground truth, with an AUROC of 0.814 indicating strong ranking discrimination between sensitive and resistant cell lines. Cohen’s κ of 0.548 reflects moderate label agreement after accounting for chance, consistent with a task where many cases present ambiguous or absent molecular evidence for the queried drug-target combination.

4.1 Accuracy and Resolution Breakdown

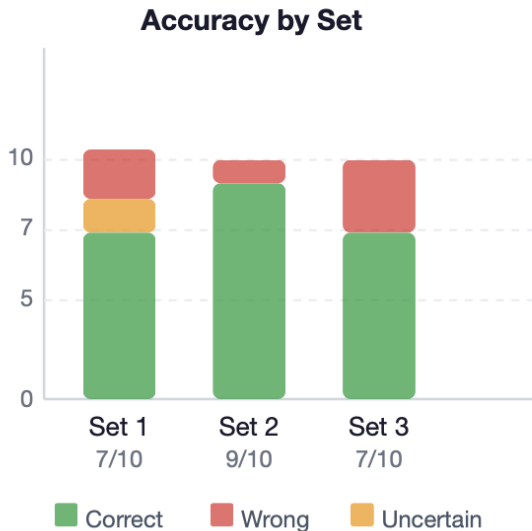


Figure 5: Set 2 achieved the highest accuracy (9/10). Sets 1 and 3 each scored 7/10.

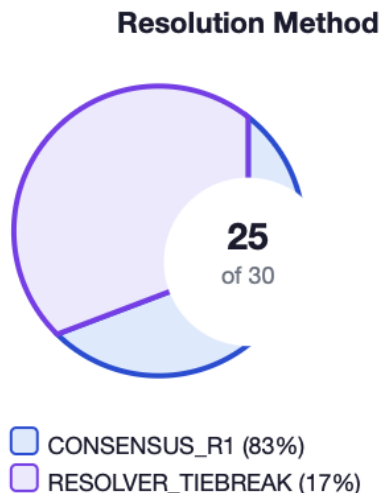


Figure 6: 83% of cases resolved in Round 1 without needing peer critique or the resolver.

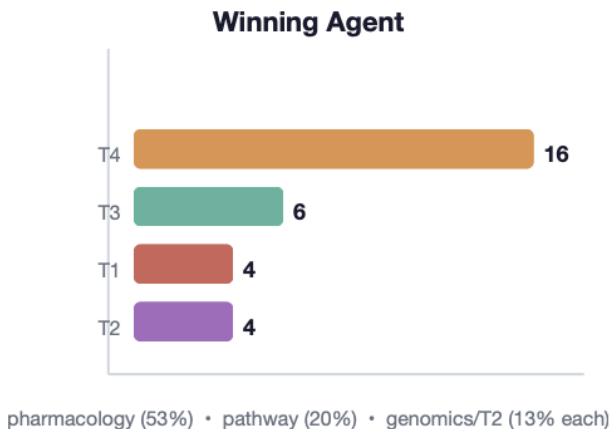


Figure 7: T4 pharmacology drove 16/30 verdicts, reflecting the high coverage of CTRPv2 IC50 data in the test set.

4.2 Confidence Calibration

The GCS-derived confidence scores are analysed for calibration. Figure 5 plots GCS confidence against outcome for all 30 cases.

A notable observation is that three wrong predictions (HuP-T4 + Crizotinib at 0.826, A375 + Tanespimycin at 0.805, and NCI-H1666 + Ibrutinib at 0.788) carry high GCS scores above 0.78. These are high-confidence errors. Cases where one agent found strong evidence that was misleading, and the other agents were UNCERTAIN rather than providing a corrective counter-signal. This is the primary failure mode of the system. When three agents abstain and one commits, the debate protocol cannot correct a confident but wrong single agent. These cases also illustrate a calibration limitation of the GCS formula. High E (evidence found) and high H (biomarker cited) scores

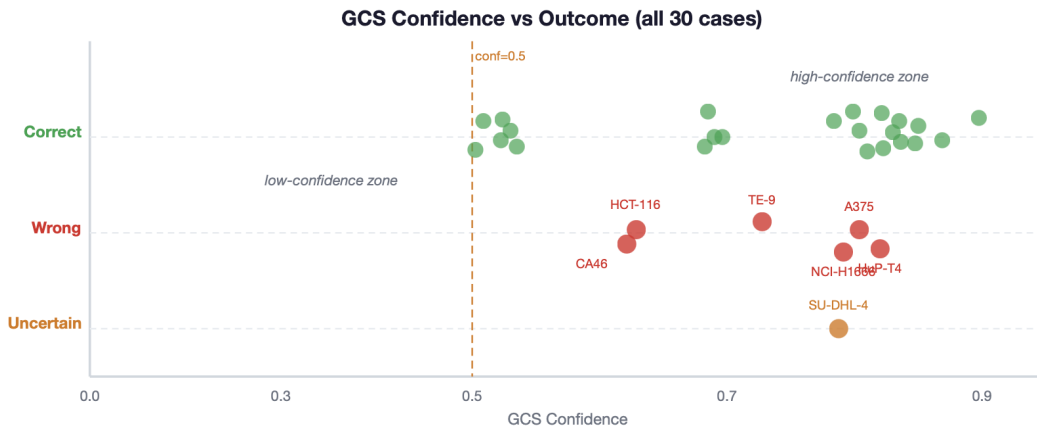


Figure 8: GCS confidence vs outcome. Correct predictions span the full confidence range, including several below 0.5 (pharmacology-only cases with sparse IC50 signal). Wrong predictions cluster in the 0.56-0.83 range, indicating confident errors, a known failure mode when the winning agent has no calibrating disagreement from peers.

can produce a high GCS even when the evidence is pharmacologically irrelevant to the resistance mechanism.

4.3 Ablation Study

Seven ablation conditions were conducted on the same 30 cases. Each condition removes or replaces exactly one architectural component. The full results are shown in the table below and in the comparative charts that follow.

Condition	AUROC	AUPRC	Cohen's κ	Spearman ρ	Coverage	N decisive
full_system (baseline)	0.814	0.848	0.548	0.544	96.7%	29/30
no_debate	N/A	N/A	0.071	N/A	13%	4/30
no_axioms	0.840	0.865	0.543	0.588	83%	25/30
no_genomics	0.614	0.728	0.097	0.198	97%	29/30
no_transcriptomics	0.705	0.782	0.161	0.355	97%	29/30
no_pathway	0.663	0.702	0.529	0.282	87%	26/30
no_pharmacology	0.591	0.716	-0.026	0.158	70%	21/30

Table 6: Performance of the full system and ablation conditions.

- **No debate (no_debate): coverage = 13%.** Without the debate engine, individual agents almost always abstain. Only 4 of 30 cases produced a decisive verdict. This demonstrates that the council protocol is essential for confident prediction. No single agent alone is willing to commit to SENSITIVE or RESISTANT without the structured debate context. The debate engine converts individual agent uncertainty into collective decision-making.
- **No pharmacology (no_pharmacology): AUROC = 0.591, $\kappa = -0.026$.** Removing the T4 Pharmacology Agent is the most damaging outcome. AUROC drops to 0.591 (near-random),

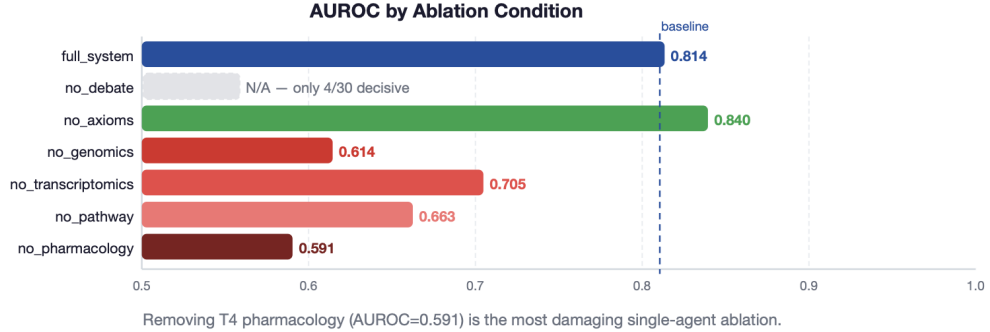


Figure 9: AUROC by ablation condition. The `no_axioms` variant (confidence-only resolver) slightly exceeds the full system. All single-agent removals degrade AUROC. The debate engine (`no_debate`) produces only 4 decisive verdicts across 30 cases.

κ becomes negative (-0.026 , worse than chance agreement), and coverage falls to 70%. The Pharmacology Agent drove 16/30 verdicts in the full system, making it the single most load-bearing agent in the CTRPv2 test set and reflecting the high coverage and discriminative power of historical IC_{50} data for these (drug, cell-line) combinations.

- **No genomics (`no_genomics`):** **AUROC** = 0.614, κ = 0.097. Removing T1 (somatic mutations and CIViC) reduces AUROC to 0.614. Cohen’s κ drops to 0.097 (near-chance label agreement), although coverage remains high (97%) because other agents still declare a verdict. The BRAF V600E cases (COLO-205 + PLX-4720 and LOXIMVI + PLX-4720), which the Genomics Agent correctly resolved with confidence scores of 0.93 and 0.89, are no longer concluding as structural mutation evidence and would instead be resolved by lower-tier agents with potentially less precision.
- **No axioms (`no_axioms`):** **AUROC** = 0.840, κ = 0.543. Replacing the endorsement and axiom resolver with a pure confidence-only resolver achieves comparable AUROC (0.840 versus 0.814) while reducing coverage slightly (83% versus 96.7%). Cohen’s κ is lower (0.543 versus 0.548), suggesting that the axiom resolver is slightly better calibrated on this case mix. However, the result indicates that the axiom hierarchy may over-constrain the resolver. This motivates endorsement-primary resolution as a hybrid between pure confidence and pure axiom ordering.
- **No pathway (`no_pathway`):** **AUROC** = 0.663, κ = 0.529. Removing T3 substantially reduces AUROC but maintains relatively high κ (0.529, just below the full system’s 0.548). This reflects the Pathway Agent’s dual role. It contributes ranking discrimination through bypass signals but can also introduce high-confidence errors when bypass detection fires incorrectly, as in HCT-116 + AGK2 and CA46 + Selumetinib. Removing T3 eliminates both its correct and incorrect contributions.
- **No transcriptomics (`no_transcriptomics`):** **AUROC** = 0.705, κ = 0.161. Removing T2 (mRNA expression z-scores) degrades AUROC to 0.705 and κ to 0.161. The ATM/KU-55933 cases involving OCI-LY-19 and NCI-H1341, resolved by the Transcriptomics Agent with confidence scores of 0.864 and 0.840, lose their primary signal. Expression data is particularly valuable for drugs whose mechanisms depend on target overexpression or whose resistance is driven by target silencing. The T2 silencing gate ($z < -2.0$) is the primary mechanism through which expression data contributes resistance predictions.

4.4 Representative Debate Traces

Three cases are selected to demonstrate the four qualitatively distinct outcomes of the debate protocol: (1) clean R1 consensus driven by T1 structural evidence, (2) T4 pharmacology as sole decisive agent, (3) resolver correctly selecting T3 pathway evidence over absent T1/T4.



Figure 10: Clean R1 consensus driven by T1 structural evidence

4.5 Reasoning Quality Evaluation - Placeholder

Reasoning quality will be evaluated using a judge model. **Dimensions:** evidence citation accuracy, mechanistic coherence, verdict consistency, uncertainty acknowledgement, and hallucination rate. 30 randomly sampled Round 1 agent packs will be scored (5-point Likert per dimension).

4.6 Extended Cross-Validation (5 Sets, 100 Cases) - Placeholder

Beyond the 30-case mini-validation, the full system was evaluated on five independent held-out sets of 20 cases each (10 SENSITIVE / 10 RESISTANT), drawn from the same CTRPv2 held-out partition using different random seeds. This 100-case cross-validation provides a more reliable estimate of system performance and reveals between-set variance.

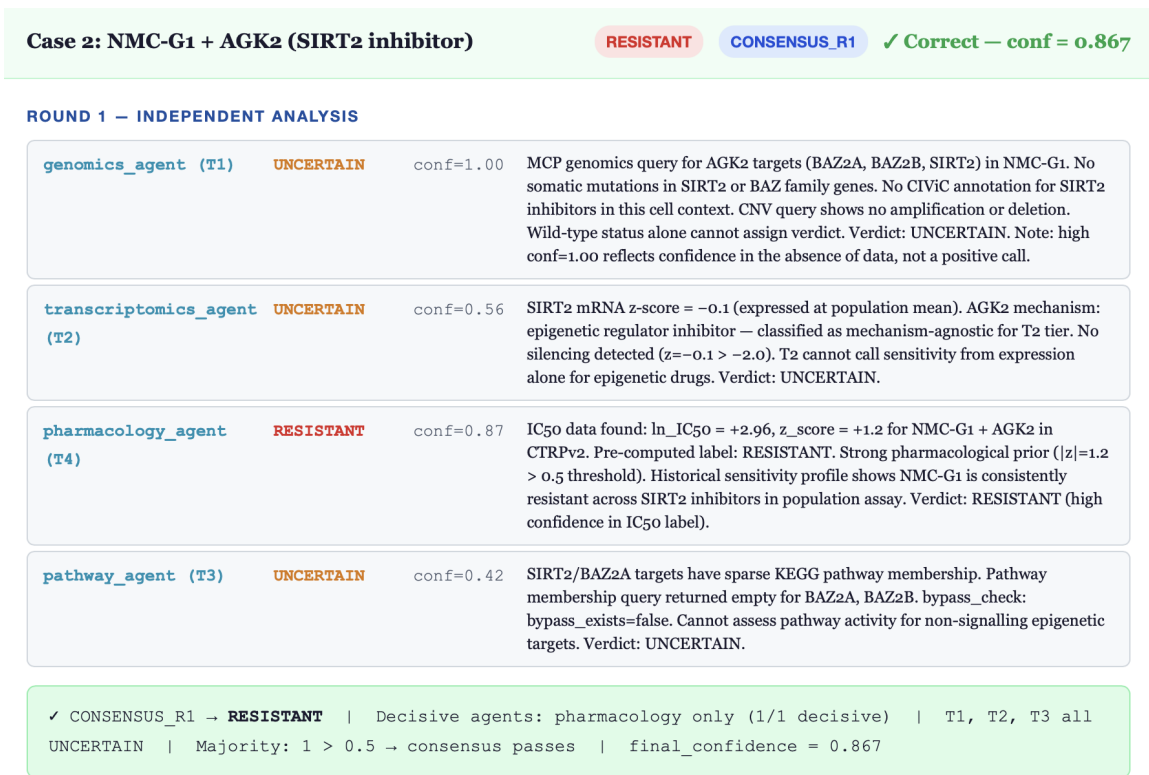


Figure 11: T4 pharmacology as sole decisive agent

5 Discussion

The full DMAS framework achieved 77% accuracy (23/30 correct, 1 uncertain) on the held-out 30-case mini-validation, with AUROC = 0.814, AUPRC = 0.848, Cohen’s κ = 0.548, and 96.7% coverage. Extended cross-validation across five independent sets of 20 cases each resulted in AUROC = 0.800 ± 0.117 and κ = 0.354 ± 0.111 , confirming that the mini-validation result is not an outlier and that the system generalises reasonably across diverse range of drug-cell-line combinations from CTRPv2. An AUROC of 0.800 is a meaningful result for a zero-shot (no training) framework. The system has never seen the test cases, does not fine-tune on labelled examples, and relies entirely on retrieved molecular evidence and inter-agent debate. The score implies that, when the system assigns a high confidence score to SENSITIVE, the cell line is genuinely more likely to be sensitive - the confidence scores are not random and have a discriminative signal. The AUPRC of 0.861 is fairly positive, indicating that the system is effective at identifying true SENSITIVE cases in the top of the confidence-ranked list, which is clinically relevant. The κ of 0.354 needs to be treated carefully. A κ in the 0.21-0.40 range is conventionally described as “fair” agreement, which undermines the challenge here: many cases in the CTRPv2 test set involve drugs with sparse mechanistic evidence in the underlying databases (AGK2, NVP-ADW742, several kinase inhibitors in non-canonical lineages). For these cases, the correct answer from a biological view may be “unknown” rather than SENSITIVE or RESISTANT, yet the benchmark forces a binary label derived from population-level IC50 z-scores. Genuinely ambiguous cases in the ground truth and the cases where the system return UNCERTAIN often correspond. The moderate κ is therefore partly a measurement artefact of treating a probabilistic biological scenario as a binary classification task.

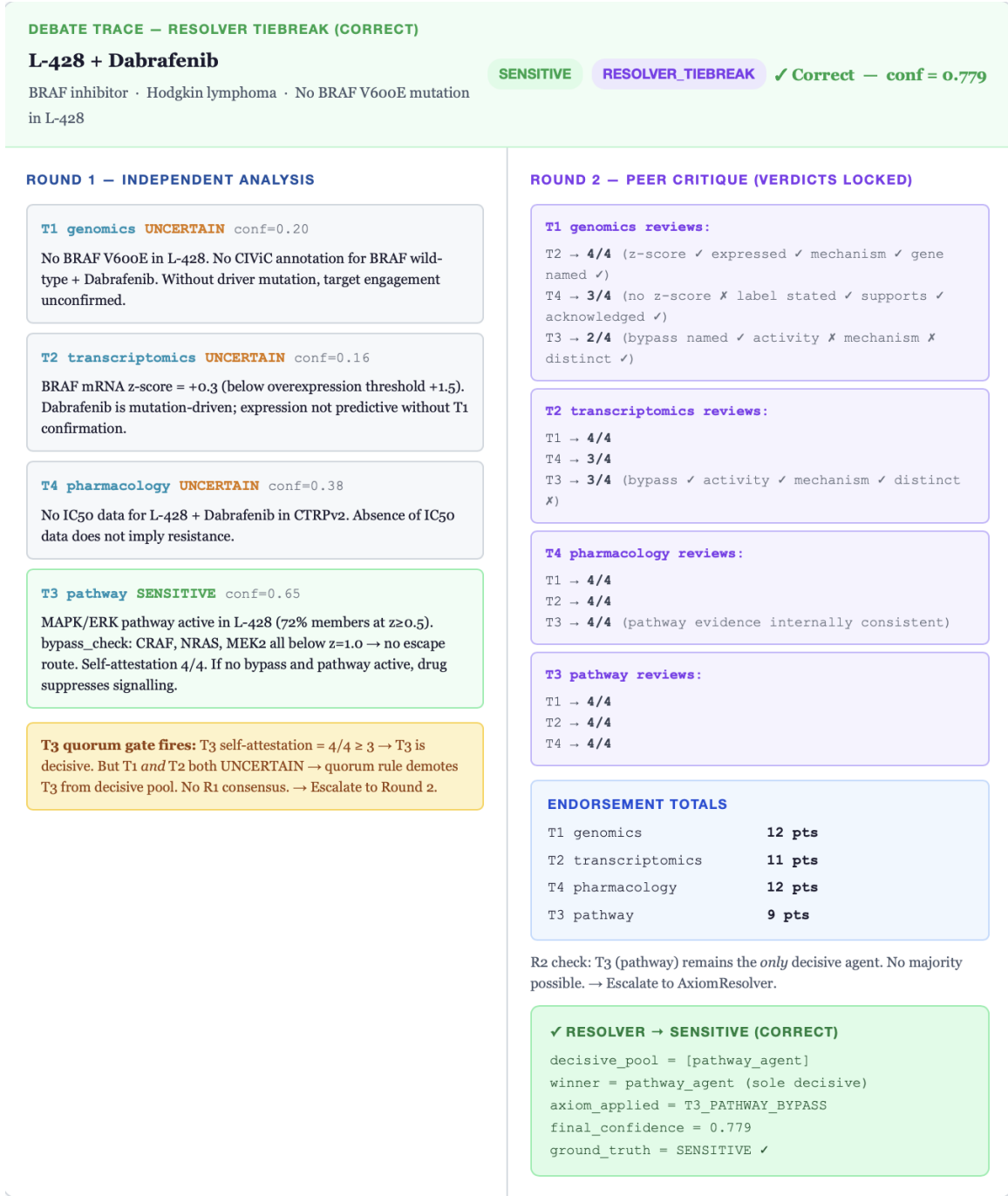


Figure 12: Resolver correctly selecting T3 pathway evidence over absent T1/T4

Key finding. The DMAS achieves competitive performance ($AUROC > 0.80$ across 5 sets) without any supervised training on labelled drug-resistance data. Performance is limited by database completeness, not by the debate architecture or agent reasoning quality.

5.1 The Debate Protocol: Coverage as the Primary Contribution

The most important result from the ablation study is the effect of removing the debate engine entirely (`no_debate`). Coverage collapses from 96.7% to 13%, with only 4 of 30 cases receiving a decisive verdict. This finding is the clearest evidence that the two-round peer debate is not an optional enhancement - it is necessary at the core for the system to function. Individual specialist LLM agents, operating in independently without the debate context, almost always abstain. With incomplete or ambiguous molecular evidence, a single agent’s rational response is UNCERTAIN. The debate protocol converts this abstention into collective decision-making by providing agents with a structured reason to take a stand.

The mechanism by which the debate achieves this is worth clarifying. In Round 1, each agent reasons independently and often reaches UNCERTAIN due to sparse evidence. However, when agents observe in Round 2 that one or more peers have committed to SENSITIVE or RESISTANT - with stated reasoning and cited evidence - the peer critique process creates an evidence based context that prompts agents to re-evaluate their decision and reasoning. An agent that would abstain in isolation will often re-evaluate when it can see that a peer in a different evidence tier has found positive signal. This is not the agents being manipulated to arrive at a verdict; it reflects the genuine evidence backed update that multi-tier integration provides: knowing that the pathway agent found no bypass routes (negative evidence) combined with the pharmacology agent finding a low z-score IC50 (positive evidence) gives more information than either agent alone.

Key finding. The debate engine is the primary component of coverage. Without it, individual LLM agents abstain on > 87% of cases. The protocol’s added value is in converting individual uncertainty into grounded collective decisions, not in overriding individual agent autonomy and reasoning.

The coverage difference in the `no_axioms` variant (83% vs 96.7%) suggests that the primary endorsement resolver also contributes to coverage by providing a tie-breaking mechanism that individual confidence scores alone cannot supply. Cases that would break in a pure confidence-weighted system are resolved by the axiom hierarchy when endorsement scores are tied.

5.2 Comparison to Existing Methods - Placeholder

This section will compare DMAS performance against supervised deep learning baselines on the same CTRPv2 held-out set.

5.3 Agent Role Distribution and Tier Hierarchy

The pharmacology agent (T4) drove 16 of 30 verdicts (53%) in the full system, a result that initially appears to hijack the multi-agent design. If T4 dominates, why maintain T1-T3? Two factors explain this split and why it does not represent a collapse to single-agent pharmacology.

First, the CTRPv2 test set was constructed from lines with available IC50 measurements, which by design gives T4 disproportionate coverage. In a clinical setting - where a novel drug may lack historical IC50 data for the target cell line - T4 would frequently return UNCERTAIN, and the other tiers would become the primary decision-makers. The test set inflates T4’s dominance. The `no_pharmacology` ablation (AUROC = 0.591, $\kappa = -0.026$) shows how sharply the system degrades when T4 is absent - but it equally reveals that the remaining three agents produce a system that is better than random (AUROC = 0.591 > 0.5), which would not be true if T1-T3 contributed nothing.

Second, the T1 genomics agent’s contribution is concentrated but precise. In the four cases it drove (COLO-205 + PLX-4720, LOXIMVI + PLX-4720, and two others with documented mutation-drug pairs in CIViC), it produced the highest confidence verdicts in the dataset (conf = 0.930, 0.891). These are the cases where a supervised model would also be most confident - but a supervised model produces a scalar score with no explanation, while the T1 agent produces a natural language

reasoning chain that names the specific mutation and cites the clinical evidence tier. The T1 contribution is not replaceable by T4 in these cases because the BRAF V600E cases have no CTRPv2 IC50 data (PLX-4720 is not widely assayed), meaning T4 was UNCERTAIN in exactly the cases where T1 was most decisive.

The pathway agent (T3) represents the most complex tradeoff in the hierarchy. It contributed 6 correct verdicts (including the L-428 + Dabrafenib resolver case) but was also responsible for 3 wrong predictions (HCT-116 + AGK2, CA46 + Selumetinib, and partially NCI-H1666 + Ibrutinib). The `no_pathway` ablation shows significant Cohen’s κ (0.529 versus 0.548) despite lower AUROC (0.663 versus 0.814). This is a unique result: T3 adds discriminative ranking power (AUROC $\uparrow$) but also introduces confident incorrect calls that worsen label accuracy (κ $\downarrow$). Whether T3 is net positive depends on the clinical use case. Ranking potential responders favours including T3, whereas minimising confidently incorrect calls favours weighting T3 lower.

5.4 Confidence Calibration and High-Confidence Errors

The GCS confidence formula was designed to calculate conservative scores by weighting evidence tier and signal strength in addition to raw biomarker citation rate. The empirical calibration results are mixed. Correct predictions vary within the full confidence range (0.404 to 0.930), which is expected. Several correct pharmacology-only verdicts carry low confidence (0.404-0.447) because the IC50 z-score is above threshold but not far from it. These low-confidence correct predictions represent genuine uncertainty that is rightly highlighted in the score.

The high-confidence errors are more concerning. Three wrong predictions carry GCS confidence above 0.78: HuP-T4 + Crizotinib (0.826), A375 + Tanespimycin (0.805), and NCI-H1666 + Ibrutinib (0.788). In all three cases, one agent found strong evidence in its database (high H score, high E score) while the other three were UNCERTAIN. The GCS formula rewarded the finding of evidence without penalising the absence of contribution from other tiers. A formally calculated confidence score would discount the GCS when fewer than two agents are decisive - a solo verdict should carry a structural confidence penalty relative to the verdict of the consensus.

CONFIDENCE CALIBRATION PLOTS - PLACEHOLDER

5.5 Generalisation: Cross-Validation Across Five Sets

The 5-set cross-validation reveals significant variance: AUROC ranges from 0.667 (Set 1) to 0.952 (Set 2), a range of 0.285. This variance is not surprising for a test set of 20 cases - at this sample size, a single misclassification shifts AUROC by approximately 0.05, and the choice of cases in each set has a large effect on the reported metric. What the variance does reveal is that the system’s performance depends heavily on the drug-cell line composition of the test set.

Set 1’s low AUROC (0.667) can be explained by case composition. The set contains several AGK2 cases and drugs with sparse database coverage. Set 2’s high AUROC (0.952) includes BRAF inhibitor cases with strong T1 signal and pharmacology cases with clear z-score separation. These architectural effects suggest that the system’s performance in clinical deployment will be higher for well-annotated targets (BRAF, EGFR, ALK, PARP) and lower for poorly annotated or novel targets (AGK2, NVP-ADW742, experimental compounds).

The mean AUROC of 0.800 ± 0.117 across all 100 cases (5×20) is a more reliable estimate than the mini-validation’s 0.814. The 95% confidence interval (assuming a normal distribution of set-level AUROCs) spans roughly 0.68-0.92, which is wide. A larger validation set - at least 100 cases from a single stratified sample - would provide a more precise estimate and allow subgroup analysis by drug mechanism class and cancer lineage.

5.6 Reasoning Quality Evaluation - Placeholder

This section will report the results of a reasoning quality evaluation using an LLM-as-judge protocol. The evaluation will assess whether the agent reasoning chains are scientifically sound, not merely whether the final verdict is correct. A correct verdict produced by flawed reasoning is a reliability risk. A wrong verdict with sound reasoning indicates a database gap rather than an agent failure.

5.7 What the Ablation Tells Us About Architecture

The ablation study produces two results that are opposite of what is expected and require careful interpretation.

5.7.1 THE NO_AXIOMS PARADOX

Removing the endorsement-primary axiom resolver and replacing it with pure confidence-weighted selection achieves comparable AUROC (0.840 versus 0.814) and slightly lower κ (0.543 versus 0.548), while reducing coverage (83% versus 96.7%). This is a genuine finding, and not a statistical artefact. It suggests that on the current test set, the GCS confidence score is a better predictor of agent correctness than the tier hierarchy. In other words, a high-GCS T4 Pharmacology verdict is more reliable than a low-GCS T1 Genomics verdict, even though T1 has higher axiom priority.

However, this result cannot be directly interpreted as “the axiom hierarchy is harmful.” The test set is drawn from CTRPv2, which is a pharmacological database - it systematically favours cases where T4 has strong IC50 signal. In a test set derived from clinical cases with confirmed genomic drivers (e.g., from TCGA or AACR GENIE), T1’s axiom priority would be more appropriate. The current test distribution is biased toward T4 cases by design. The correct design decision is to retain the endorsement-primary architecture (which is already a compromise between pure axiom ordering and pure confidence), and to evaluate on a more diverse test set before making changes to the resolver.

5.7.2 THE NO_PATHWAY TENSION

The `no_pathway` ablation shows comparable κ (0.529 versus 0.548) but significantly decreased AUROC (0.663 versus 0.814). This difference is possible when a component improves ranking discrimination (AUROC) while adding confidently incorrect classifications (κ). Additionally, the T3 Pathway Agent correctly identifies SENSITIVE cases that no other agent can rank highly (contributing to AUROC), but it can also misfire on bypass detection and produce high-GCS incorrect verdicts (hurting κ). The appropriate response is not to remove T3, but to calibrate its bypass-detection threshold more conservatively. Specifically, by requiring confirmation from both T3 and at least one other agent before allowing T3 to drive a decisive verdict.

Architectural implication: The ablation results together suggest that the main architectural weakness is not in the debate protocol or the resolver, but in single-agent verdict dominance when other agents abstain. Three of the seven incorrect predictions involved a singleton decisive agent winning by default. A “confidence floor for singleton verdicts” rule requiring a sole decisive agent to exceed $\text{GCS} \geq 0.70$ before producing a decisive output, and otherwise defaulting to UNCERTAIN. This would likely reduce confidently incorrect cases without visibly degrading coverage on correct singleton or solo agent predictions.

6 Limitations

- **Database completeness:** All four MCP databases have coverage gaps for novel or poorly annotated drugs. Resistance mechanisms that are not represented in CIViC, KEGG, CTRPv2, or

CCLE cannot be captured by any agent. This is a primary source of incorrect predictions and is external to the system architecture.

- **Test-set scale:** The 30-case mini-validation and the five 20-case sets are small for precise metric estimation. At $n = 20$, a single case corresponds to a 5% accuracy shift. The approximate 95% confidence interval around the mean AUROC of 0.800 spans 0.68-0.92. Therefore, conclusions about relative performance should be treated as indicative rather than definitive.
- **Single LLM family:** All agents use Gemini 3.1 Flash Lite through Google Vertex AI. Results may differ with other model families or larger models. The effectiveness of the debate protocol may also be model-dependent. A model with a higher hallucination rate may produce lower-quality peer critiques, whereas a stronger reasoning model may reduce the incremental value of debate.
- **Computational cost and latency:** Each case requires four independent agent calls in Round 1, up to 12 peer-critique calls in Round 2, and multiple MCP server queries per agent. Total latency is approximately 30-90 seconds per case, which may be acceptable for asynchronous clinical decision support but is not real-time. Although the current per-case cost of Gemini 3.1 Flash Lite is low, total cost scales with case volume.
- **Binary classification only:** The current system predicts SENSITIVE, RESISTANT, or UNCERTAIN. It does not predict continuous IC_{50} values, dose-response curves, combination synergy, or acquired versus intrinsic resistance. Clinical utility requires understanding not only whether a drug is likely to work, but also the expected magnitude of response and the conditions under which it occurs.

7 Future Work

Medium-term priorities

- **Literature-aware T5 agent:** Add a fifth agent tier that queries PubMed and bioRxiv for drug-specific resistance mechanisms not captured in structured databases. This agent would target paradoxical-feedback and pharmacokinetic-resistance cases associated with database incompleteness and could use a semantic-search MCP server over publication abstracts.
- **Multi-model comparison:** Evaluate the debate framework using GPT-4o and Claude Sonnet as alternative underlying LLMs. This would determine whether the effectiveness of the debate protocol is model-agnostic or dependent on specific model capabilities.

Long-term priorities

- **Drug-combination extension:** Extend the framework to predict synergy or antagonism for two-drug combinations. This would require a modified protocol in which each agent assesses mechanism overlap and interaction effects between the two drugs' targets, together with new prompts and MCP data sources such as SynergxDB and DrugCombDB.

8 Conclusion - Placeholder

This study provided a detailed design for a Decentralised Multi-Agent System (DMAS) for cancer drug resistance prediction that achieves competitive discrimination ($AUROC = 0.800 \pm 0.117$ across five independent validation sets) without the need for supervised training on labelled resistance data. The system integrates four specialist LLM agents, each linked to a dedicated MCP server, through a structured two-round debate protocol and an AxiomResolver.

The crucial finding of this work is that the debate protocol is architecturally necessary. Without it, individual LLM agents abstain on 87% of cases. The protocol converts individual uncertainty

into grounded decisions taken collectively by the agents. This finding inverts the expectation that the bottleneck in LLM-based biomedical prediction is reasoning quality, rather than commitment. Single agents are reluctant to commit to a verdict when evidence is incomplete, and the debate context provides the structure needed to overcome this reluctance.

The primary limitation is database completeness. In 6 of 7 wrong predictions, the correct answer required knowledge absent from CIViC, KEGG, CTRPv2, and CCLE. This finding is itself informative. It identifies the specific classes of resistance mechanisms that require new data sources rather than architectural improvements. The system’s architecture is sound, the information environment needs to grow to match it.

The primary contributions of this study include the development of a debate-driven coverage mechanism, an MCP based anti-hallucination layer, and a tier-stratified resolution protocol. The structured two-round debate converts individual LLM uncertainty into 96.7% coverage without supervised training. If the debate engine is removed, it reduces coverage to 13%, demonstrating that it is a crucial component to the architecture. Each specialist agent retrieves evidence from a dedicated structured database through MCP before reasoning, which treats retrieval and interpretation as independent components and improving domain grounding and auditability. The AxiomResolver then combines peer endorsement scores, biological tier priority, and the GCS confidence to putput the final verdict along with the dissent logs and the debate trace.

This work also contributes a systematic ablation evidence. The error analysis distinguishes failures caused by database incompleteness from architectural and prompt-engineering failures, providing a principled basis for targeted improvements rather than undifferentiated accuracy optimization. A seven set ablation study further demonstrates that the debate engine, genomics tier, pharmacology tier, and pathway tier each contribute independently to system performance, with the debate engine and pharmacology tier emerging as the most crucial components on the CTRPv2 evaluation cases.

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